

Karthhek Vikash Meesala

Network Engineer

Texas, USA | Ph: (361)246-2993 | Mail: Meesalakarthheekvikash@gmail.com | LinkedIn: [Karthhek Vikash](#)

PROFESSIONAL SUMMARY:

- 6+ years of experience as a Network Engineer with a proven track record in designing, implementing and managing complex network infrastructures across diverse industries including healthcare, finance and e-commerce.
- Expertise in architecting secure & compliant network solutions incorporating Next-Gen Firewalls (Cisco ASA, Palo Alto, Fortinet), SD-WAN (Cisco Viptela, VMware VeloCloud), 802.1X authentication, VLAN segmentation, SIEM platforms (Splunk, IBM QRadar), ITSM frameworks (ServiceNow, Jira Service Management) & automation tools (Ansible, Python, Terraform).
- Proficient in network monitoring (SolarWinds, PRTG, Nagios), deep packet analysis (Wireshark, NetFlow, sFlow), QoS implementation (Cisco NBAR, F5, NetScaler), AAA services (TACACS+), VPN technologies (Check Point) and cloud networking (Google Cloud, Azure, AWS).
- Adept at ensuring regulatory compliance (HIPAA, HITRUST, NIST, RBI, ISO 27001, PCI DSS) and optimizing network performance for critical applications.

TECHNICAL SKILLS:

Network Security:	Cisco ASA, Palo Alto, Fortinet NGFW, Palo Alto Panorama, Check Point, Web Application Firewall (WAF)
Routing & Switching:	Cisco Catalyst, Arista Switches, OSPF, BGP, VLAN, Inter-VLAN Routing
SD-WAN:	Cisco Viptela, VMware VeloCloud
Cloud Networking:	AWS VPC, AWS Direct Connect, Azure Firewall, Google Cloud (GCP)
Automation & Scripting:	Ansible, Terraform, Python
Monitoring & Performance:	SolarWinds, PRTG, Nagios, NetFlow, sFlow, Wireshark
SIEM & Log Analysis:	Splunk, IBM QRadar
Identity & Access Management:	802.1X Authentication, TACACS+, AAA
VPN & Secure Access:	Check Point VPN, Cisco AnyConnect
Quality of Service (QoS):	Cisco NBAR, F5, NetScaler
Network Infrastructure:	MPLS, FIX Protocol, SD-WAN, RFID, POS System
Compliance & Security Standards:	HIPAA, HITRUST, NIST, PCI DSS, ISO 27001, RBI Guidelines
IT Service Management (ITSM):	ServiceNow, Jira Service Management
Testing & Troubleshooting:	Wireshark, NetFlow, sFlow
Enterprise Collaboration:	Jira, Confluence

PROFESSIONAL EXPERIENCE:

Cigna Healthcare – TX | August 2024 – Present

Network Engineer

- Designed and implemented a HIPAA-compliant network architecture, integrating Next-Gen Firewalls (Cisco ASA, Palo Alto) and reducing security breaches by 35%, ensuring compliance with HITRUST and NIST security frameworks.
- Deployed SD-WAN solutions (Cisco Viptela, VMware VeloCloud) to optimize remote healthcare facility connectivity, achieving a 40% reduction in bandwidth costs while improving telemedicine service quality.
- Engineered 802.1X authentication with VLAN segmentation, strengthening network access control and ensuring secure, role-based access for electronic health records (EHR) and medical devices.
- Integrated SIEM platforms (Splunk, IBM QRadar) for real-time threat detection, enhancing incident response efficiency by 45% and improving forensic analysis for healthcare security audits.
- Developed and enforced ITSM frameworks (ServiceNow, Jira Service Management) to streamline network incident tracking, change management and service requests, reducing resolution times by 30%.
- Automated network configurations using Ansible and Python scripting, eliminating 60% of manual interventions, minimizing human errors and ensuring consistent configurations across multiple healthcare sites.
- Proactively monitored network performance using SolarWinds, PRTG and Nagios, identifying and resolving network bottlenecks that reduced downtime by 30% and improved clinician access to patient records.
- Performed deep packet analysis with Wireshark, NetFlow and sFlow, optimizing network troubleshooting and reducing mean time to resolution (MTTR) by 25%, ensuring uninterrupted healthcare application availability.
- Enhanced Quality of Service (QoS) and bandwidth allocation using Cisco NBAR, F5 and NetScaler, ensuring low-latency network performance for telemedicine, imaging systems and EHR applications, reducing congestion by 20%.
- Implemented Infrastructure as Code (IaC) with Terraform, enabling rapid network provisioning and automation within Google Cloud (GCP) hybrid environments, accelerating infrastructure deployment by 50%.

Zensar Technologies – India | January 2021 - December 2023

Network Engineer

- Architected a hybrid SD-WAN solution using Cisco Viptela, enabling secure connectivity for branch offices to the central data center; achieved a 30% reduction in MPLS operational costs through optimized routing protocols.
- Deployed Fortinet Next-Generation Firewalls (NGFWs) and Palo Alto Panorama, enhancing centralized firewall management across all banking branches; boosted threat detection capabilities by 40%.
- Optimized FIX protocol-based market data feeds by implementing low-latency Arista switches; reduced trade execution times by 20%, ensuring faster financial transaction processing.
- Designed and implemented VLAN configurations and inter-VLAN routing using Cisco Catalyst switches to segregate sensitive banking data traffic securely.
- Configured OSPF and BGP routing protocols to ensure seamless communication between branch offices and the central data center, maintaining high availability and reliability of financial services.
- Conducted comprehensive security audits aligned with Reserve Bank of India (RBI) guidelines and ISO 27001 standards, ensuring compliance across all network infrastructure.
- Integrated SWIFT connectivity with advanced firewall rules, securing international financial transactions against potential cyber threats.
- Deployed Azure Firewall and Kubernetes CNI plugins, enabling secure containerized banking applications for cloud-native environments.
- Implemented AAA services using TACACS+, facilitating role-based access control across critical network devices to meet regulatory standards for data security.
- Monitored network performance using SolarWinds, achieving 99% SLA compliance for uptime; proactively identified and resolved bottlenecks in real-time.
- Integrated Splunk SIEM for real-time log analysis, enabling rapid incident response and improving overall cybersecurity posture across banking systems.
- Configured secure VPN tunnels using Check Point gateways, providing encrypted remote access for employees while maintaining compliance with banking security policies.

BitInfoCom Technologies – India | January 2019 - December 2021

Network Engineer

- Spearheaded the deployment of MPLS and SD-WAN technologies, achieving a remarkable 30% reduction in latency, thereby enhancing the user experience for millions of online shoppers.
- Diligently maintained PCI DSS compliance across all network operations, safeguarding the integrity of over millions payment transactions monthly and ensuring adherence to industry security standards.
- Expertly configured and managed advanced firewall solutions (Cisco ASA, Palo Alto) to fortify the network perimeter, effectively protecting sensitive customer data from potential breaches.
- Designed and implemented robust network monitoring frameworks utilizing SolarWinds and Nagios, enabling real-time visibility and proactive management of network performance and health.
- Leveraged automation tools such as Ansible and Terraform to streamline network configurations, resulting in a 40% reduction in deployment time and enhancing operational efficiency.
- Partnered with IT, logistics and operations teams to seamlessly integrate RFID and barcode scanning technologies into the Point of Sale (POS) system, optimizing inventory management and transaction accuracy.
- Established and maintained secure VPN solutions for remote employees, ensuring reliable and encrypted connectivity to corporate resources, thereby supporting a flexible work environment.
- Managed and optimized cloud networking solutions on AWS, including the configuration of Virtual Private Clouds (VPC) and Direct Connect, to enhance scalability and performance for e-commerce applications.
- Developed and enforced QoS policies to prioritize critical e-commerce traffic, ensuring optimal performance during peak shopping periods and minimizing downtime.
- Integrated and managed web application firewalls (WAF) to defend against prevalent web threats, enhancing the security posture of e-commerce platforms and protecting customer information.

EDUCATION:

Masters in Computers & Information sciences - Texas A&M University, TX, USA

Bachelor of Mechanical Engineering - Aditya Institute of Technology and Management College Andhra Pradesh, India

Certifications:

- AWS Solution Architect Professional